Rational Inattention to Equilibrium Objects

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Motivation

- Traditional rational inattention: pay attention to some exogenous random variable, e.g. minimize quadratic loss $(a - x)^2$ where $x \sim F(x)$ by choosing a with cost of attention to x.
- But how to pay attention to something whose value depends on your attention to it?
- Here: consumer pays attention to a distribution of prices set by monopolist, where the firm's optimal price depends on consumer's attention to it
- Consumer trades off cost of attention vs. increase in accuracy of demand, but endogeneity of price distribution to attention means their attention also changes price levels
- This introduces a new incentive for attention – it changes the equilibrium distribution of prices

General set-up

1. **Consumer**
   - Single consumer with quadratic utility, choose quantity to demand $q^*(\hat{p})$ based on perceived price $\hat{p}$, but pays true price p
   - Conjectures that distribution of prices is Gaussian, so optimal signal is Gaussian $s = p + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2) \implies$ attention is choice of $\xi \equiv \frac{\sigma_p^2}{\sigma_p^2 + \sigma_\epsilon^2}$
   - Faces cost of attention $C(\xi)$ and knows that parameters of price distribution p^d and σ_p^2 depend on attention ξ, since the distribution is an equilibrium object
2. **Monopolist:**
   - Seeks to maximize profit $\pi = (p - \tilde{c})\,\mathbb{E}_s\left[q^*(\hat{p})\right]$ with exogenous random cost $\tilde{c} \sim \mathcal{N}(\mu_c, \sigma_c^2)$
   - Consumer's quadratic utility + Gaussian conjecture of prices $\implies$ monopolist's optimal price distribution is indeed Gaussian with mean μ_p and variance σ_p^2 (interesting things happen when $p^d \neq \mu_p$; more later)
3. Results in a tractable attention problem where consumer's attention choice trades off:
   1. Costs of attention (traditional)
   2. Benefits of reducing mistakes in demand from imperfect perception of price (traditional)
   3. Benefits from lowering mean of price distribution (new)

Timing of model

1. Consumer conjectures distribution of prices is $\mathcal{N}(p^d, \sigma_p^2)$ and since utility is quadratic, optimal signal about price is of form $s = p + \epsilon$ for $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ where they choose attention $\xi = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_\epsilon^2}$
2. Consumer chooses level of attention ξ anticipating how monopolist's price distribution depends on ξ
3. Monopolist observes level of ξ and sets their optimal price given random cost draw $\tilde{c}$
4. Consumer makes purchase decision given level of attention ξ and price p

Consumer's attention problem

- Quadratic utility + conjecture of Gaussian prices $\implies$ consumer chooses attention ξ which forms perception of price after signal as $\mathbb{E}[p|s] = \hat{p} = \xi \cdot s + (1-\xi)p^d$ where $s = p + \epsilon$
- Experienced utility is $u(q^*(\hat{p}), p) = \theta q^*(\hat{p}) - \frac{(q^*(\hat{p}))^2}{2} - pq^*(\hat{p})$ where $q^*(\hat{p}) = \theta - \hat{p}$
- Since conjecture of Gaussian prices $\implies$ optimal for monopolist to post Gaussian prices, then expected utility of consumer can be written in terms of moments of said distribution. When $p^d = \mu_p$, this gives:

$$\mathbb{E}[u(\xi)] = \mathbb{E}\left[u(q^*(p)) - \frac{1}{2}(\hat{p} - p)^2\right] = \frac{1}{2}\left[(\theta - \mu_p)^2 + \xi\sigma_p^2\right] \tag{1}$$

- Therefore consumer's problem is to maximize expected utility less cost of attention

$$\max_{\xi \in [0,1]} \frac{1}{2}\left[(\theta - \mu_p)^2 + \xi\sigma_p^2\right] - C(\xi) \tag{2}$$

- To see how μ_p and σ_p^2 depend on ξ, we need to solve monopolist's problem

Monopolist's pricing problem

- Firm wishes to maximize profit over expected signal, i.e. $\max_p (p - \tilde{c})\, \mathbb{E}_s[q^*(\hat{p}(p))] = (p - \tilde{c})\left(\theta - \left(\xi p + (1-\xi)p^d\right)\right)$, thus optimal price follows

$$p^* = \frac{\tilde{c}}{2} + \frac{\theta - (1-\xi)\, p^d}{2\xi} \tag{3}$$

- If we impose that the consumer's prior $p^d = \mu_p$, then this implies that the equilibrium distribution of prices is

$$p^* \sim \mathcal{N}\left(\frac{\xi\mu_c + \theta}{1+\xi}, \frac{\sigma_c^2}{4}\right) \tag{4}$$

- Two key features: i) mean price μ_p decreasing in attention ξ (assuming $\theta > \mu_c$) ii) variance of prices not changed by ξ

Consumer's attention with optimal prices

- FOC for consumer's problem (knowing that $\mu_p(\xi)$ but σ_p^2 independent of ξ) gives:

$$\underbrace{(\mu_p(\xi) - \theta)\frac{d\mu_p(\xi)}{d\xi}}_{\text{benefit from lower mean price}} + \underbrace{\frac{\sigma_p^2}{2}}_{\text{more attention when prices volatile}} - \underbrace{C'(\xi)}_{\text{cost of attention}} = 0 \tag{5}$$

- Now that we know how μ_p depends on ξ (and that $\sigma_p^2 = \sigma_c^2/4$, we can write the solution with Shannon Entropy cost $C(\xi) = -\frac{\lambda}{2}\log(1-\xi)$ gives

$$\frac{\xi}{(1+\xi)^3} \cdot (\theta - \mu_c)^2 + \frac{\sigma_c^2}{8} - \frac{\lambda}{2}\frac{1}{1-\xi} = 0 \tag{6}$$

- At $d\mu_p/d\xi = 0$, this reduces to traditional solution: $\xi^* = \max\left\{0, 1 - \frac{\lambda}{\sigma_p^2}\right\}$, but since $d\mu_p/d\xi < 0$, ξ^* is higher. As usual, $d\xi^*/d\sigma_p^2 > 0$ and $d\xi^*/d\lambda < 0$. Also, ξ^* increasing in size of price adjustment

Moving beyond rational expectations

- Previous treatment assumed consumer has correct prior mean, i.e. $p^d = \mu_p$. This implies that $d\mu_p/d\xi < 0$ always
- If we allow for 'sticky priors', i.e. $p^d \neq \mu_p$, the problem is richer
  1. Consumer benefits from higher attention ξ because distortionary effects of incorrect prior are lessened
  2. However, response of price to more attention is richer: we can have $d\mu_p/d\xi > 0$ with sufficiently high p^d!
- Why can $d\mu_p/d\xi > 0$? If expected price p^d is very high, monopolist has to price low to compensate – increasing ξ reduces role of high p^d, relieving monopolist of 'bad reputation', allowing them to price higher
- Next step: re-do model, allowing $p^d = \mu_p + \varsigma(\xi)$ where $\varsigma(\xi)$ is the distortion in their prior mean

Consumer's attention problem with sticky priors

- Now that $p^d \neq \mu_p$, $\mathbb{E}\left[(\hat{p}-p)^2\right]$ is not just related to variance of prices and attention – bias is introduced with distortion of prior $\varsigma(\xi)$; this changes consumer's utility:

$$\max_{\xi\in[0,1]} \mathbb{E}\left[u(q^*(\hat{p}),p^*(\xi),p^d)\right] = \frac{1}{2}\left[(\theta-\mu_p)^2+\xi\sigma_p^2\right] - \frac{1}{2}\left[(1-\xi)^2\left(\varsigma(\xi)\right)^2\right] - C(\xi) \quad (7)$$

- Terms in black constitute problem from before, but new term reflects that $\varsigma(\xi) = p^d - \mu_p$ is the gap between the prior and the true mean price
- Again need to see how distribution of price μ_p and σ_p^2 depend on ξ – turns out variance is still invariant to attention, but the mean is more interesting

Monopolist's pricing problem with sticky consumer priors

- Optimal price formula is same as before, but $p^d \neq \mu_p$ means that distribution of prices is

$$p^* \sim \mathcal{N}\left(\frac{\mu_c + p^d}{2} + \frac{\theta - p^d}{2\xi}, \frac{\sigma_c^2}{4}\right) \tag{8}$$

- Key features are
  1. Mean price is decreasing in expected price p^d:

$$\frac{d\mu_p}{dp^d} = \frac{\xi - 1}{2\xi} < 0 \tag{9}$$

  2. Mean price can be *increasing or decreasing* in attention (price increasing in attention if expected price is very high, i.e. $p^d > \theta$)

$$\frac{d\mu_p}{d\xi} = \frac{p^d - \theta}{2\xi^2} \tag{10}$$

  - Higher p^d forces monopolist to price lower to compensate for 'bad reputation'
  - Higher ξ means less weight on p^d
  - For sufficiently high p^d, higher ξ relieves monopolist of bad reputation $\Longrightarrow$ price *higher*

Consumer's attention with sticky priors and optimal prices

- Optimal attention now solves

$$\underbrace{(\mu_p(\xi) - \theta)\frac{d\mu_p(\xi)}{d\xi}}_{\text{change in mean price}} + \underbrace{\frac{\sigma_p^2}{2}}_{\text{more attention when prices volatile}} + \overbrace{(1-\xi)(\varsigma(\xi))^2}^{\text{reduced influence of distorted prior}} - \overbrace{(1-\xi)^2\varsigma(\xi)\frac{d\varsigma}{d\xi}}^{\text{change in distortion w/ more attention}} - \underbrace{C'(\xi)}_{\text{cost of attention}} = 0$$

- Higher ξ decreases utility losses from given distortion, but ξ changes distortion too
- Higher ξ can increase or decrease price (depending on p^d) so consumer may face trade-off between attention and price level